

Savannah L. Ferretti

✉ savannah.ferretti@uci.edu | 🌐 savannahferretti.github.io | 🔗 savannahferretti | 🐙 savannahferretti

EDUCATION

Ph.D. in Earth System Science , University of California, Irvine	Jun 2026
M.S. in Earth System Science , University of California, Irvine	Dec 2023
B.S. in Earth & Atmospheric Sciences , Cornell University	Dec 2020

EXPERIENCE

Graduate Student Researcher Sep 2021–Jun 2026
Pritchard & Baldwin Labs, University of California, Irvine Irvine, CA

- Developed statistical & machine learning models to analyze variability in South Asian monsoon rainfall using large-scale reanalysis & satellite datasets
- Quantified uncertainty & evaluated predictive skill across model configurations for robustness
- Built open-source, scalable Python workflows for parallel processing in HPC environments
- Communicated findings through peer-reviewed publications & conference presentations

Consulting Analyst Mar–Aug 2021
Higher Education Research Services, Huron Consulting Group New York, NY

- Built Excel-based financial planning & reporting tools for a research nonprofit
- Developed long-term resource forecasts & presented recommendations to executive stakeholders

Undergraduate Research Assistant Jun–Aug 2020
Commene Lab, Lamont-Doherty Earth Observatory Palisades, NY

- Quantified changes in atmospheric methane concentrations during the NYC COVID-19 shutdown
- Applied meteorological analysis & transport modeling to attribute emissions to likely source regions

SKILLS

Python (PyTorch, PySR, Dask, Xarray, Pandas, Cartopy, Matplotlib) • Bash • Git/GitHub • High-Performance/Distributed Computing • NetCDF/HDF5/Zarr • Linux • Microsoft Excel (Certified Specialist) • $\text{\LaTeX}$ • WordPress

SELECTED PUBLICATIONS

- S. L. Ferretti**, J. Lin, S. Shamekh, J. W. Baldwin, M. S. Pritchard, & T. Beucler. (2026). **Data-driven integration kernels for interpretable nonlocal operator learning**. In press at *Environ. Data Sci.*
- S. L. Ferretti**, M. S. Pritchard, F. Ahmed, L. Peng, & J. W. Baldwin. (2025). **Explaining South Asian monsoon rainfall seasonality using a metric of plume buoyancy**. *Geophys. Res. Lett.*
- S. Yu, W. M. Hannah, L. Peng, ...**S. L. Ferretti**, ...& M. S. Pritchard. (2023). **ClimSim: A large multi-scale dataset for hybrid physics–ML climate emulation**. *NeurIPS 2023 Datasets & Benchmarks Track*, Outstanding Paper Award.